

2018 Ph H2 Q4

Section: Our Dynamic Universe

Topic: Projectile motion, friction, relative velocity

A car is driven up a ramp on one moving lorry, attempting to land on the roof of another moving lorry, and then stop while that lorry is still moving.

Question: Using your knowledge of physics, comment on the challenges involved in carrying out the stunt successfully.

Worked solution

- Relative velocity: The second lorry must be moving at the correct speed relative to the car's horizontal velocity so that the car lands on it rather than behind or in front.
- Projectile motion: The car's trajectory after leaving the ramp must be carefully calculated. The horizontal distance covered during flight must match the gap between lorries.
- Ramp launch: The angle and speed at which the car leaves the ramp determine the car's projectile path. Too steep or too shallow will miss the target.
- Vertical motion: Gravity causes the car to fall while airborne, so timing of motion must ensure the lorry roof is at the right vertical position when the car arrives.

- Stopping: After landing, the car must come to rest relative to the moving lorry. This requires sufficient friction between tyres and roof surface to decelerate safely.
- Safety: Small errors in speed, ramp angle, or lorry separation could lead to the car overshooting or missing the lorry entirely.

Final answer:

The stunt requires precise control of launch speed and ramp angle, accurate alignment of the lorries' speeds and positions, and sufficient friction to allow the car to stop on landing. Any miscalculation could cause the car to miss or slide off.

Revision tips

- In stunt physics, relative motion between vehicles is crucial.
- Projectile motion depends on initial speed, angle, and gravity.
- Friction is needed to stop relative motion after landing.
- Analyse horizontal and vertical motion separately.
- Real-world uncertainties make such stunts highly risky.